Distribution-Preserving QIM

Trading noise margin for statistical undetectability and scaling robustness

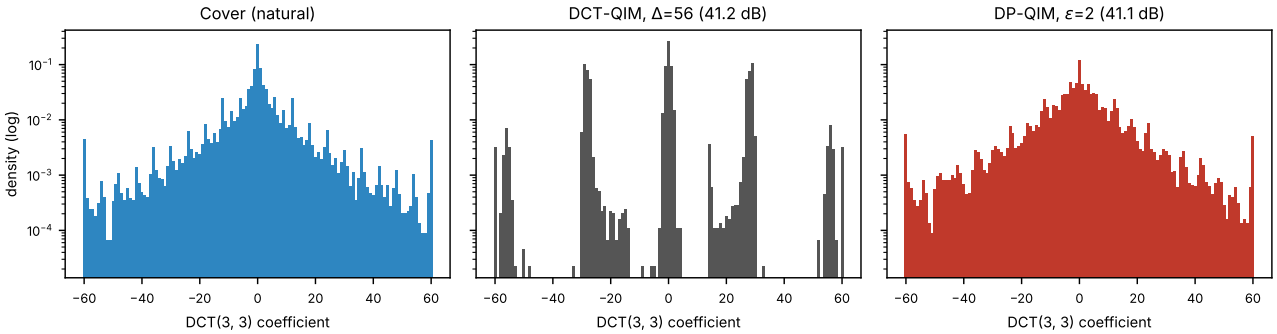
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github.com/sairam782/Invisible-Watermarking

The problem

QIM watermarks snap DCT coefficients onto a lattice, so the coefficient histogram turns into isolated spikes. Natural images have no such spikes, which lets a detector, or a black-box removal attack such as UnMarker (IEEE S&P 2025), find marked images without the key.

The idea: DP-QIM

Split the natural distribution of one DCT coefficient at its median into two equal-mass cells labelled 0 and 1. A coefficient already in the right cell is left alone (about half of them). Otherwise it is replaced by a *real* coefficient value drawn from the other cell, away from the boundary. The marked histogram is then $12F + 12F_\epsilon$, close to natural. Because the median of an AC coefficient is 0, decoding is a sign test, so scaling the coefficient by any $s > 0$ (blur, downscaling, contrast) cannot flip a bit.



Results (44 real crops, 15 photos, same DCT position, matched PSNR, 95% source-bootstrap CI)

Detectability (ROC-AUC, 0.5 = undetectable)

	Learned detector		Peak statistic	
Bits (PSNR)	DCT-QIM	DP-QIM	DCT-QIM	DP-QIM
64 (53.9)	0.90 [0.78, 1.00]	**0.52 [0.49, 0.57]**	0.72 [0.58, 0.88]	**0.49 [0.47, 0.49]**
256 (47.2)	1.00 [1.00, 1.00]	**0.61 [0.54, 0.71]**	0.86 [0.71, 1.00]	**0.47 [0.41, 0.50]**
1024 (41.1)	1.00 [0.99, 1.00]	**0.77 [0.61, 0.98]**	0.99 [0.98, 1.00]	**0.59 [0.46, 0.73]**

Robustness: DP-QIM minus matched QIM (pp)

Attack	DP-QIM	Δ vs. DCT-QIM (pp)	95% CI
No attack	0.999	-0.1	[-0.3, +0.0]
JPEG Q=90	0.839	**-15.2**	[-19.2, -11.8]
JPEG Q=70	0.710	**-28.4**	[-31.4, -25.3]
Gaussian noise σ=0.03	0.829	**-13.6**	[-16.0, -11.6]
Gaussian blur r=1	0.837	**+34.6**	[+30.3, +39.6]
Downscale 0.5×	0.831	**+34.0**	[+30.5, +37.4]
Contrast ×0.5	0.967	**+19.6**	[+13.5, +30.0]
Contrast ×0.8	0.975	-0.3	[-2.5, +2.0]
UnMarker-style (τ=2, j=0.5)	0.972	**-2.8**	[-5.2, -1.0]
UnMarker-style (τ=1.5, j=1)	0.912	**-6.9**	[-9.6, -3.9]

Take-away

Wins: near-chance detectability; +20 to +35 pp under blur, downscaling and contrast reduction. **Costs:** −14 to −28 pp under JPEG and additive noise, and 3 to 7 pp worse under an UnMarker-style removal attack. Hiding the watermark's statistics is a real lever, but it does not by itself defeat removal. The released protocol (source-grouped CV, matched PSNR, independent attacker, bootstrap CIs) also documents five evaluation pitfalls that changed our own conclusions.